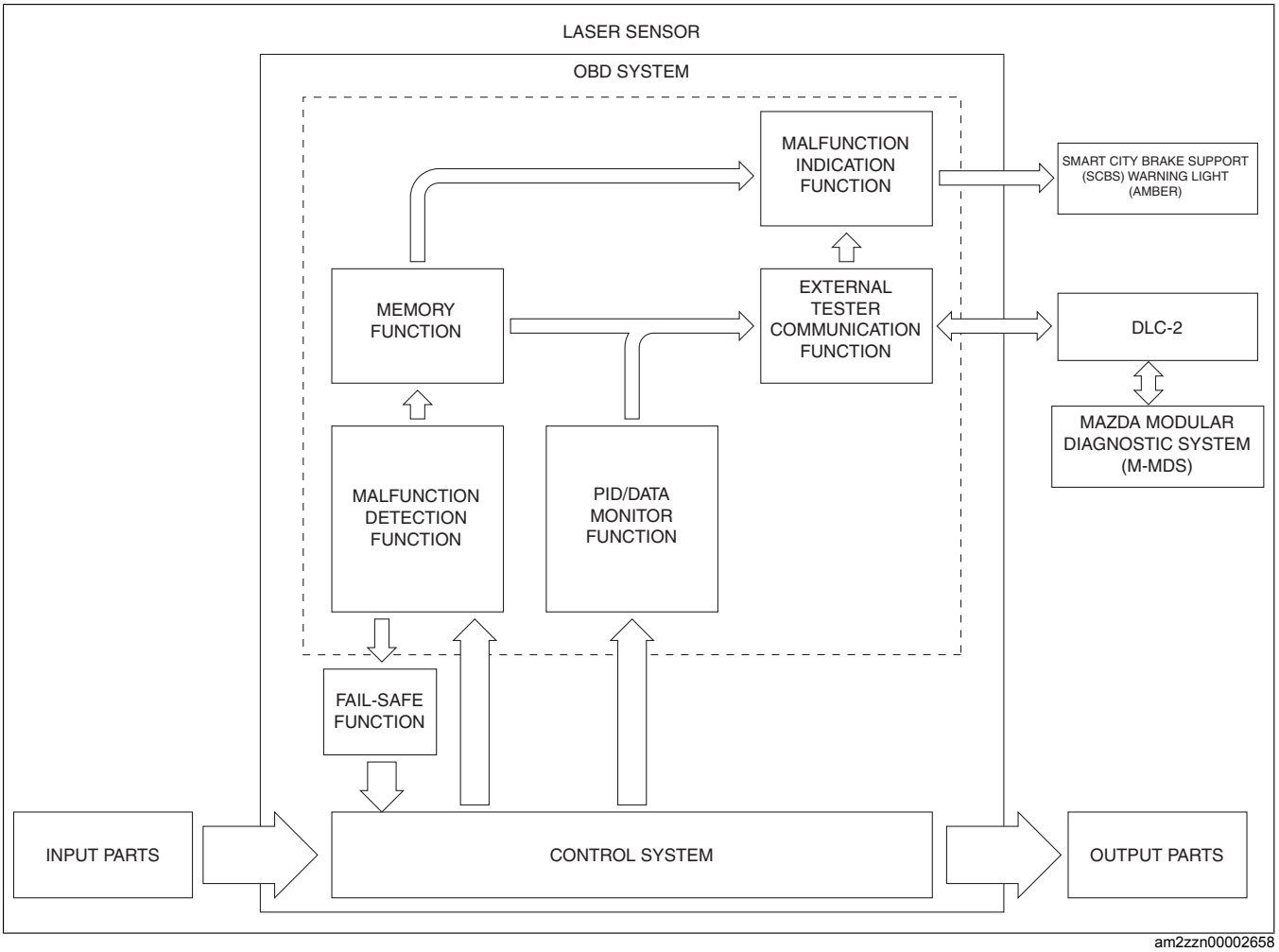
ON-BOARD DIAGNOSTIC SYSTEM [LASER SENSOR]

id0402c4181300

Outline

- The on-board diagnostic system consists of the following functions:
  - — A malfunction detection function that detects abnormalities in input/output signals when the ignition switch is at the ON position.
  - — A PID/data monitor function that reads out specified input/output signals.
- The data link connector 2 (DLC-2), which groups together all the connectors used for malfunction diagnosis into a single location, has been adopted, thereby improving serviceability. Diagnosis is performed by connecting the Mazda Modular Diagnostic System (M-MDS) to the DLC-2.
- In addition to DTC read-out, the Mazda Modular Diagnostic System (M-MDS) has following functions:
  - — PID/data monitor access
  - — DTC clear using the display screen of the diagnostic tester

Block diagram



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Function

Malfunction detection function

- The malfunction detection function detects malfunctions in the input/output signal system of the laser sensor when the ignition switched ON or driving the vehicle.

Malfunction display function

- When the malfunction detection function detects a malfunction, the Smart City Brake Support (SCBS) warning light (amber) illuminates to alert the driver.
- Using the external tester communication function, DTCs can be output to the DLC-2 via the CAN line. At the same time, malfunction detection results are sent to the memory function and fail-safe function.

Memory function

- The memory function stores DTCs of malfunctions in input/output signal systems. With this function, once a DTC is stored it is not cleared after the ignition switch has been switched OFF, even if the malfunctioning signal system has returned to normal.

- Since the laser sensor has a built-in non-volatile memory, DTCs are not cleared even if the battery is removed. Therefore, it is necessary to clear the memory after performing repairs. Refer to the Workshop Manual for the DTC clearing procedure.

DTC table

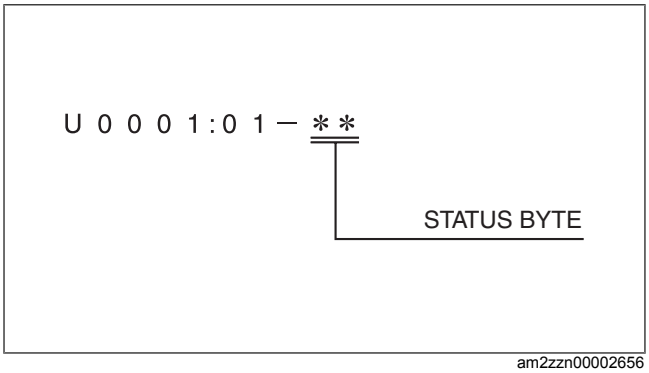
×: Applicable
—: Not applicable

DTC	Smart City Brake Support (SCBS) warning light (amber)	Description	Fail-safe	Drive cycle	Self test type*1	Memory function
U0001:00	Illuminates	CAN line	×	—	C	×
U0100:00						
U0121:00						
U0131:00						
U0155:00						
U0401:00	Illuminates	Signal error from PCM	×	—	C	×
U0401:82						
U0415:00	Illuminates	Signal error from DSC HU/CM	×	—	C	×
U0415:68						
U0415:82						
U0420:00	Illuminates	Signal error from EPS control module	×	—	C	×
U0420:82						
U0423:00	Illuminates	Signal error from instrument cluster	×	—	C	×
U0423:82						
U1A14:49	Illuminates	Laser sensor	×	—	C	×
U2300:54	Illuminates	Laser sensor configuration	×	—	C	×
U2300:55						
U2300:56						
U2300:64						
U3000:00	Illuminates	Laser sensor (internal malfunction)	×	—	C	×
U3000:64						
U3000:66	Illuminates	Smart City Brake Support (SCBS) system	×	—	C	×
U3003:16	Not illuminate	Battery power supply	×	—	C	×
U3003:17						

*1 : C: CMDTC self test

Status byte for DTC

- The status byte is the two-digit code (two digits after hyphen (-)) after the seven-digit DTC.
- The status byte is a code which indicates the pending code, current/past malfunction status, or warning illumination status.
- The status byte is a code which indicates the pending code, current/past malfunction status, or warning illumination status.
- The status byte can be read by performing a CMDTC self test using the Mazda Modular Diagnostic System (M-MDS).
- For details on the status byte, refer to the explanation on the M-MDS when reading the DTC.



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Fail-safe function

- When the malfunction detection function determines a malfunction, the Smart City Brake Support (SCBS) warning light (amber) illuminates to alert the driver. At this time, the fail-safe function controls as indicated in the fail-safe function table.

Fail-safe function table

DTC	Smart City Brake Support (SCBS) control
U0001:00	Control disabled
U0100:00	Control disabled
U0121:00	Control disabled

DTC	Smart City Brake Support (SCBS) control
U0131:00	Control disabled
U0155:00	Control disabled
U0401:00	Control disabled
U0401:82	Control disabled
U0415:00	Control disabled
U0415:68	Control disabled
U0415:82	Control disabled
U0420:00	Control disabled
U0420:82	Control disabled
U0423:00	Control disabled
U0423:82	Control disabled
U1A14:49	Control disabled
U2300:54	Control disabled
U2300:55	Control disabled
U2300:56	Control disabled
U2300:64	Control disabled
U3000:00	Control disabled
U3000:64	Control disabled
U3000:66	Control disabled
U3003:16	Control disabled
U3003:17	Control disabled

Snapshot data

- The snapshot data stores the currently detected DTC data.

—: Not applicable

Snapshot data item	Unit	Definition	Data read/use method	Corresponding data monitor items
AAT	°C, °F	Ambient air temperature	—	—
APP_STATUS	Accelerator Pedal Off/ Under 20%/ Over 20%/ FAIL	Accelerator pedal position	—	—
CFG_STATUS	Config Complete/ Not Configured/ Config Error	Configuration status	—	—
ECT_STATUS	Under 0 degrees C/ 0 - Under 80 degrees C/ Over 80 degrees C/ FAIL	Engine coolant temperature status	—	—
IC_VPWR	V	Instrument cluster power supply	• The laser sensor constantly receives the power supply voltage value of the instrument cluster sent via CAN signal from the instrument cluster. • If a DTC is detected, the laser sensor records the power supply voltage of the instrument cluster when the DTC was detected, and it is displayed in the M-MDS.	VPWR*1

Snapshot data item	Unit	Definition	Data read/use method	Corresponding data monitor items
IG-ON_TIMER	hh:mm:ss*2	Elapsed time since ignition was switched ON **Note** • The instrument cluster records the elapsed time since the ignition was switched ON.	• The laser sensor constantly receives the elapsed time since the ignition was switched ON sent via CAN signal from the instrument cluster. • If a DTC is detected, the laser sensor records the elapsed time since the ignition was switched ON when the DTC was detected, and it is displayed in the M-MDS.	—
PWR_MODE_KEY	Key Out/Key Recently Out/Key Approved (Position 0)/Post Accessory (Position 0)/ Accessory (Position 1)/ Post Ignition (Position 1)/Ignition On (Position 2)/Running (Position 2)/ Running - Starting In Progress (Position 2)/ Crank (Position 3)	• Key Out: Ignition switched off • Key Recently Out (Position 0): Elapsed time within 3 s since ignition was switched off • Accessory (Position 1): Ignition is switched to ACC • Post Ignition (Position 2): Elapsed time within 3 s since ignition was switched ON • Ignition On (Position 2): Ignition switched ON (engine off) • Running (Position 2): Ignition switched ON (engine on) • Running - Starting: Cranking condition	• The laser sensor constantly receives the ignition switch status sent via CAN signal from the instrument cluster. • If a DTC is detected, the laser sensor records the ignition switch status when the DTC was detected, and it is displayed in the M-MDS.	—
RPM_STATUS	Engine Stop/ Under 1500 rpm/ Over 1500 rpm/ FAIL	Engine RPM status	• The laser sensor constantly receives the ignition switch status sent via CAN signal from the instrument cluster. • If a DTC is detected, the laser sensor records the ignition switch status when the DTC was detected, and it is displayed in the M-MDS.	TACHOMTR*1

Snapshot data item	Unit	Definition	Data read/use method	Corresponding data monitor items
SHIFT_STATUS	P/N D/ R/ FAIL	Shift position status	• The laser sensor constantly receives the selector lever position sent via CAN signal from the instrument cluster. • If a DTC is detected, the laser sensor records the selector lever position when the DTC was detected, and it is displayed in the M-MDS.	—
TOTAL_DIST	km, ft, mi	Accumulated total traveled distance from completion of vehicle until laser sensor detects DTC (Odometer value in instrument cluster)	The distance traveled when the laser sensor detected a DTC can be calculated by performing the following procedure. 1. Verify the odometer value in the instrument cluster. 2. Verify the snap shot data item TOTAL_DIST. 3. Subtract 2 from 1.	—
TOTAL_TIME	hh:mm:ss*2	Accumulated total elapsed time since vehicle completion until laser sensor detects a DTC **Note** • When the ROOM fuse is removed, and the ignition is switched off, the time is not included in the elapsed time.	The elapsed time when the laser sensor detected a DTC can be calculated by performing the following procedure. 1. Verify the PID item TOTAL_TIME of the instrument cluster. 2. Verify the snap shot data item TOTAL_TIME. 3. Subtract 2 from 1.	TOTAL_TIME*1
VPWR	V	Power supply	—	VPWR_IG1
VSPD_STATUS	Stop/ 0 - 10 km/h/ Over 10 km/h/ FAIL	Vehicle speed status	• The laser sensor constantly receives the vehicle speed sent via CAN signal from the instrument cluster. • If a DTC is detected, the laser sensor records the vehicle speed when the DTC was detected, and it is displayed in the M-MDS.	SPEEDOMTR*1

*1 : PID for instrument cluster
*2 : Fractions of seconds may be indicated after the decimal point

PID/data monitor function

- The PID/data monitor function is used for optionally selecting input/output signal monitor items preset in the laser sensor and reading them out in real-time.

Monitor item (Mazda Modular Diagnostic System (M-MDS) display)	Definition	Unit/operation
DST_BMP_TGT	Distance from front bumper to target that laser sensor has detected	m, ft
VPWR_IG1	Power supply	V
VSPD	Vehicle speed	KPH, MPH

External tester communication function

- The external tester communication function enables communication of diagnostic data (DTC read-outs, input/output signal read-outs) between the laser sensor and an external tester.
- By connecting the Mazda modular diagnostic system (M-MDS) to the DLC-2, diagnostic data can be sent and received between the Mazda modular diagnostic system (M-MDS) and the laser sensor via CAN lines.
- The laser sensor receives the command signals of the malfunction detection function and PID/data monitor function from the Mazda modular diagnostic system (M-MDS), and sends DTCs and data regarding the operating status of each input/output part to the Mazda modular diagnostic system (M-MDS).

Diagnostic function name	Signal received	Signal sent
Malfunction detection function	DTC verification signal	DTC(s)
PID/data monitor function	Command signal to read selected monitor item	Monitored data for requested monitor item